# Worksheet #17

*Week 9, Day 1: k-NN and decision trees*

Name: _____
NetID: _____
Date: _____

## Guided checkpoints

For full credit, fill in the checkpoints. If you skip a checkpoint, you must include your own notes for full credit.

### 1. Talk to Your Neighbors

Fill in the Self column for yourself. Then talk to five classmates (at most one of them can be in your project group) and fill in each cell with 1 if they have taken the course or 0 if they have not.

| Course Name: | Self | Neighbor 1 | Neighbor 2 | Neighbor 3 | Neighbor 4 | Neighbor 5 |
|---|---|---|---|---|---|---|
| CS 100 Discovering Computer Science | | | | | | |
| CS 251 Data Structures | | | | | | |
| CS 301 Languages and Automata | | | | | | |
| CS 361 Systems Programming | | | | | | |
| CS 401 Computer Algorithms I | | | | | | |
| CS 411 Artificial Intelligence I | | | | | | |
| CS 412 Introduction to Machine Learning | | | | | | |
| CS 424 Visualization and Visual Analytics | | | | | | |
| CS 426 Video Game Design and Development | | | | | | |
| CS 427 Creative Coding | | | | | | |
| CS 480 Database Systems | | | | | | |
| CS 488 Introduction to Cryptography | | | | | | |

### 2. Binary Similarity

Compare your Self column with each neighbor from checkpoint 1. Count $N_{11}$ (both took it), $N_{10}$ (only you), $N_{01}$ (only your neighbor), and $N_{00}$ (neither), then compute Jaccard and SMC.

| | $N_{11}$ | $N_{10}$ | $N_{01}$ | $N_{00}$ | Jaccard | SMC |
|---|---|---|---|---|---|---|
| Neighbor 1 | | | | | | |
| Neighbor 2 | | | | | | |
| Neighbor 3 | | | | | | |
| Neighbor 4 | | | | | | |
| Neighbor 5 | | | | | | |

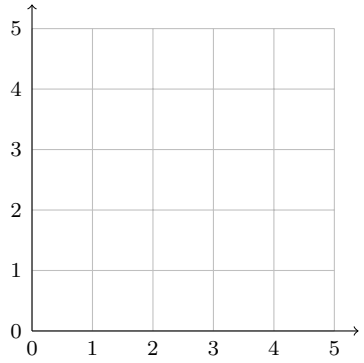
Nearest neighbor by Jaccard: _____ Nearest neighbor by SMC: _____

### 3. Jaccard vs. SMC

What is the difference between Jaccard and SMC?

### 4. Distance Metrics

Plot two points, then draw and label the Euclidean and Manhattan paths.




Euclidean:

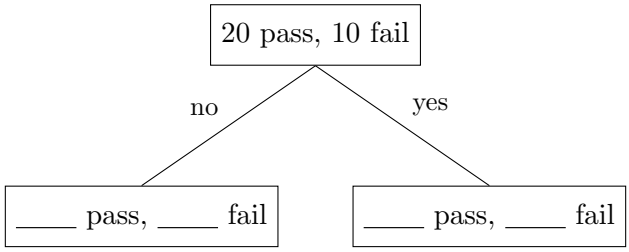
Manhattan:

### 5. Memorizing vs. Generalizing

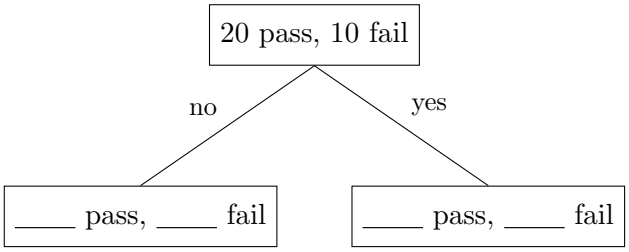
In your own words, what is the difference between memorizing and generalizing?

### 6. Good Split, Bad Split

Good split:




Bad split:




### 7. How Much Does a Split Help?

What metric(s) can be used to determine how much a split helps distinguish classes?